KENYA MEDICAL TRAINING COLLEGE
FACULTY OF REHABILITATIVE SCIENCES
DEPARTMENT OF MEDICAL ENGINEERING

FINAL QUALIFYING EXAMINATION
FOR
CERTIFICATE IN MEDICAL ENGINEERING TECHNOLOGY

PAPER: ELECTRICAL ENGINEERING THEORY

DATE: 17th June 2022 **TIME: 3 HOURS (9:00AM – 12:00 PM)**

INSTRUCTIONS

1. This paper consists of:
   - Section 1 (40 Multiple Choice Questions)
   - Section 2 (8 Short Answer Questions)
   - Section 3 (1 Long Answer Question)
2. Attempt **ALL** Questions
3. Write the **EXAMINATION NUMBER** given on all the answer sheets provided and on the question paper.
4. Ensure that all examination answer scripts are handed in at the end of the examination
5. Ensure you sign the examination register provided

EXAMINATION NUMBER C/MET/JUN-22/FAE/868

ELECTRICAL ENGINEERING

SECTION 1: MULTIPLE CHOICE QUESTIONS (40 MARKS)

1. The armature resistance of DC is 0.5, the supply voltage is 200v and back EMF of 196v at full speed. Armature current is?
   a) 4A
   b) 8A
   c) 400A
   d) 392A
2. A commutator is a device fitted to a DC generator so as to
   a) Convert AC into placating DC
   b) Provide carbon brushes
   c) Produce flus
   d) Increase voltage
3. The base requirement of a DC machine, armature winding is that it must be?
   a) Closed one
   b) Wave
   c) Lap
   d) Open winding
4. The armature of a DC machine is laminated to
   a) Reduce the bulk
   b) Provide the bulk
   c) Insulate eddy currents
   d) Reduce eddy currents
5. In a lap winding the number of brushes is always
   a) Double the number of poles
   b) Same as the number of poles
   c) Half the number of poles
   d) Two
6. In a DC generator the generated EMF is directly proportional to the
   a) Field current
   b) Armature conductors
   c) Field pole winding
   d) Armature current
7. Compensating windings are used in D.C generators
   a) Mainly to reduce the eddy currents and provide local short circuits
   b) To provide part for circulation of casting air
   c) To neutralize the cross magnetism effect of armature reaction
   d) None of the above

8. Branches are physically placed in the
   a) Magnetic natural axis
   b) Geometric natural axis
   c) Both magnetic and geometric neutral axis
   d) Interpoles
9. The effect of inserting a resistance in series with the field winding of a shunt motor is
   a) Increase the magnetic field
   b) Increase the speed of the motor
   c) Decrease the armature current
   d) Increase the field current
10. The shaft torque of a motor is less than armature torque because
   a) Copper losses
   b) Iron losses
   c) Rotational losses
   d) Hysteresis losses
11. In small AC generators
   a) The magnetic field is produced by DC electromagnets placed on the stationery member
   b) The current induced in the rotating armature is collected by means of slip rings
   c) Armature is the revolving member
   d) All of the above
12. Which one of the following types of alternators is used for in hydroelectric power station?
   a) None salient pole
   b) Turbo-generator
   c) Salient pole
   d) Steam turbine
13. The maximum speed at which an alternator can be driven to generate 50Hz and 400V having 2 poles is
   a) 400 RPM
   b) 3600 RPM
   c) 3000 RPM
   d) 1500 RPM
14. Three phase alternators are invariably star connected because
   a) Magnetic losses are minimized
   b) Less number of turns of wire required
   c) Smaller conductors can be used
   d) Higher terminal voltage is obtained
15. While running, a synchronous motor is compelled to run at synchronous speed because
   a) Damper windings in its field winding
   b) Magnetic lock between stator and rotor poles
   c) Induced E.M.F in the rotor field winding by stator flux
   d) Compulsion due to Lenz's Law

16. The direction of a rotation of a synchronous motor are used to
    a) Improve power factor
    b) Eliminate hunting of the rotor
    c) Minimize temperature of the rotor
    d) Minimize temperature rise
17. In a three phase squirrel cage induction motor
    a) Rotor conductors re short-circuited through end rings
    b) Rotor conductors end are short circuited through slip rings
    c) Rotor conductors are kept open
    d) None of the above
18. In order to operate, a single phase induction motor requires
    a) Slip rings
    b) Brushes
    c) Commutator
    d) Starting system
19. The capacitor in a capacitor start induction motor is connected in series with the _____winding
    a) Start
    b) Run
    c) Squirrel cage
    d) Compensating
20. Transformer action requires
    a) Constant magnetic flux
    b) Increasing magnetic flux
    c) Alternating magnetic flux
    d) Alternating electric flux
21. For the same rating a.d.c machine has _____an a.c machine
    a) Less weight than
    b) The same weight as
    c) More weight than
    d) Equal weight as
22. Which formula is used to calculate the generated voltage in a generator?
    a) $E_g = V - I_a R_a$
    b) $E_g = V + I_a R_a$
    c) $E_g = \frac{V + I_a R_a}{\emptyset}$
    d) $E_g = E_b + I_a R_a$
23. DC series motors are used in these applications where ____ _____ is required
    a) High torque
    b) Constant speed
    c) Low-no-load speed
    d) High no-load speed

24. In a DC machine, maximum losses occur due to_____
    a) Mechanical losses
    b) Iron losses
    c) Copper losses
    d) Friction and windage losses
25. The iron-core is used to _____ of the transformer
    a) Increase the weight
    b) Provide high magnetic coupling
    c) Reduce core losses
    d) Increase magnetic flux
26. Which of the following statements is **not** true about induction motors?
    a) Induction motors work on the transformer principal
    b) Induction motor operate on a slip
    c) The rotor speed and magnetic field speed are equal
    d) The rotor is the rotating part of an induction motor
27. If the slip of a 3-phase induction motor increases the power factor of the rotor circuit_____
    a) Remains unchanged
    b) Is decreased
    c) Is increased
    d) Shows no positive response
28. Which one among the following is not a single phase motor?
    a) Capacitor start motor
    b) Capacitor run motor
    c) Stepper motor
    d) Shaded pole motor
29. Alternators driven by _____ do not have tendency to hunt
    a) Water turbines
    b) Diesel turbines
    c) Steam turbines
    d) Air turbines
30. When a number of alternators are operating in parallel the power factor at which each operates is determined by_____
    a) Its field excitation
    b) Driving torque of the prime mover
    c) Power factor of the load
    d) Supply frequency
31. In a DC machine, the number of commutator segments is equal to _____
    a) Twice the number of poles
    b) Number of conductors
    c) Number of coils
    d) Number of brushes

40. The following are methods of improving commutation except?
    a) Using interpoles
    b) Resistance commutation
    c) Emf commutation
    d) Increasing magnetic field

SECTION 2: SHORT ANSWER QUESTIONS (40 MARKS)

41. A 230V series motor draws 50A from the supply. The resistances of the armature and series field windings are 0.2Ω and 0.1Ω respectively. Calculate:
    a) Brush voltage (1 mark)
    b) Back EMF (2 marks)
    c) Power wasted in the armature (2 marks)
42. Calculate the speed and open-circuit line and phase voltages of a 4pole 3phase 50Hz star connected alternator with 36 slots and 30 conductors per slot. The flux per pole is 0.05wb and winding factor is 1 (5marks)
43. i) Name the two types of rotors used in induction motors. (2 marks)

    ii) A 4 pole 3phase induction motor operates from a supply whose frequency is 50Hz. Calculate a synchronous speed with which the magnetic field of the stator is rotating and the speed of the rotor. (3 marks)
44. i) A transformer is the most efficient electrical machine. What are the main loses in a transformer? (3 marks)

    ii) Name any two types of transformers? (2 marks)
45. State five advantages of a three-phase induction motor? (5 marks)
46. State five methods used in starting a three-phased induction motor (5 marks)
47. What do we mean by skewed slots of a squirrel cage motor? (3 marks)
48. a) What cause mechanical losses in a d.c motor (2 marks)

    b) What are the effects of armature reaction in D.C machines. (3 marks)

SECTION 3: LONG ANSWER QUESTIONS (20 MARKS)

49.
    i. State and describe five parts of a DC machine. (10 marks)

    ii. Draw and label the diagram of capacitor start capacitor run single phase induction motor. (5 marks)

    iii. State five applications where a capacitor start capacitor run single phase induction motor are used. (5 marks)